

SECI1143 – PSDA
EXERCISE – TOPIC 2 (DATA DESCRIPTION)

QUESTION 1

Table below shows the height measurement collected from students who are finishing their primary school. Using the information provided, answer the following questions.

Height, h	Frequency, f
156.5 – 159.5	5
159.5 – 162.5	13
162.5 – 165.5	12
165.5 – 168.5	10
168.5 – 171.5	21
171.5 – 174.5	9

- a. Fill in the following table.

Height, h	Frequency, f	Mid point	$f_i X_i$	Cummulative Frequency
156.5 – 159.5	5			
159.5 – 162.5	13			
162.5 – 165.5	12			
165.5 – 168.5	10			
168.5 – 171.5	21			
171.5 – 174.5	9			

- b. Calculate the **median** for this grouped data.
c. Calculate the **mean** for this grouped data
d. Calculate the **mode** for this grouped data.

QUESTION 2

Table following shows the CGPA of a class of 15 students who undergoes the semester with fully online classes.

2.90	3.23	3.35
3.03	3.25	3.37
3.03	3.33	3.50
3.04	3.34	4.00
3.05	3.35	4.00

Table: CGPA of students.

- e. Identify the following:
 - i. 40th percentile
 - ii. The percentile for 3.5
- f. Calculate the following:
 - i. Inter-Quartile Range
 - ii. 2nd Quartile
- g. Calculate the following:
 - i. Mean
 - ii. Median
- h. Which measurement will be better in justifying the performance of the students in this semester? Why?

QUESTION 3

Mommy would like to have a birthday celebration for her daughter, Iman. During her survey, she purchased two bags of chocolate bars. There are 19 and 23 of chocolate bars in the first and second bags. She is interested to study the distribution of the weight (in grams) of chocolate bars that she bought. Then she asked Iman to mixed out and weighed the two bags of chocolate together. Table 1 shows the weight recorded.

Table 1

19.8	20.3	16.8	19.6	17.5	19.4	15.4
18.9	17.5	20.8	18.8	15.5	17.8	18.0
15.4	17.6	15.7	18.4	17.3	16.7	15.6
14.2	15.9	16.8	15.2	15.0	16.0	16.4
15.4	16.0	16.1	17.0	16.0	15.1	14.0
15.8	15.7	16.6	15.2	15.5	14.9	15.3

a) Complete the table below.

Class Limit	Class Boundaries	Midpoint	Tally	Frequency
14.0 – 14.9				
15.0 – 15.9				
16.0 – 16.9				
17.0 – 17.9				
18.0 – 18.9				
19.0 – 19.9				
20.0 – 20.9				

b) Based on the information in (a), **construct a histogram** of the weights of chocolate bars on the graph paper.

QUESTION 4

Mrs. Linda distributes numbers of cotton balls to each of her students. Their students received the cotton balls based on the given below:

18	13	26	12	20	14	21	16	17
22	17	25	13	30	13	22	15	21
18	18	16	21	18	31	15	19	10

a) Construct a **stem-and-leaf** plot for the given dataset.

b) Find the following of cotton balls received for each student.

- i. Range
- ii. Mode
- iii. Median